

Chapter 11 CRT Questions

1. Can data in memory be called a file? Explain.

No, memory cannot be named as a file. This is because memory holds data in a state that's temporary. A file is a series of data that indefinitely stores its own data in hard drives, SSDs, and USB drives.

2. Write the import statement required to access the File Class in an application.

In order to access the file class in an application you must use
'Import java.io.File; '

3. Identify the error in the following statement:

File textFile = new File("c:\inventory.txt");

Double backslashes would be mandatory to identify the file path because a single backslash is a wrong escape sequence in a string.

4. a) Which statement is used to write an exception handler?

The use of the try-catch-finally statement

b) Write an exception handler to handle an IOException if a specified file name cannot be used to create a file. The exception handler should display appropriate messages to the user.

```
try
{
    File file = new File(fileName);
    if (file.createNewFile())
    {
        System.out.println("File created successfully: " + fileName);
    }
    else
    {
        System.out.println("File already exists or Error");
    }
}
catch (IOException err)
```

```
System.out.println("Error: Could not create the file '" + fileName);  
System.err.println("IOException: " + err.getMessage());  
}
```

5. **a) What is the name of the stream for displaying error messages?**

System.err

b) Where are these messages displayed?

In the console

6. **a) What does the file stream keep track of?**

The file stream records and analyzes the location of the file as well as the most latest reading or writing.

b) What characters together make up a line Terminator?

The line feed character (Lf) and the carriage return character (Cr)

7. **What two classes are used together to write data to a file?**

The java.io package's FileWriter and BufferedWriter classes

8. **Write a statement to convert account balances that have been read from a text file to a double value and add them to totalBalance.**

```
totalBalance += Double.parseDouble(accountBalance);
```

9. **Explain the difference between object serialization and object deserialization.**

Object serialization is the process of writing objects to a file by converting an object's class information into a stream. Object deserialization is the process of retrieving objects from a file.

10. What interface must be implemented if objects of a class are to be written to a file?

The java.io package's Serializable interface needs to be implemented.
must implement the java.io.Serializable.